

SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY(AUTONOMOUS), CHITTOOR	
DEPARTMENT OF HUMANITIES AND SCIENCES	
Question Bank	
Name of the subject: PROBABILITY & STATISTICS	Course: II B. Tech II semester
Code: 23AHS21	
Name of the faculty: Dr. K. KOAMALA	Designation: HOD &Professor, HAS

UNIT-No. 1 Descriptive Statistics							
UNIT- N0.1	Q. No.	2 Marks	Marks	Question Type	Subject code	CO	BL
	1.	Define rank correlation.	2	SHORT	23AHS21	CO1	L1
	2.	Write the equation of regression line X on Y .	2	SHORT	23AHS21	CO1	L1
	3.	Define Primary and secondary data	2	SHORT	23AHS21	CO1	L2
	4.	Define Population and Sample	2	SHORT	23AHS21	CO1	L2
	5.	Define mean. How is it calculated for ungrouped data?	2	SHORT	23AHS21	CO1	L2
	6.	State one difference between mean and mode.	2	SHORT	23AHS21	CO1	L2
	7.	What is the formula for the mean of a frequency distribution.	2	SHORT	23AHS21	CO1	L2
	8.	State the formula to calculate the median for a grouped frequency distribution.	2	SHORT	23AHS21	CO1	L2
	9.	Write one advantage of using median over mean in data analysis.	2	SHORT	23AHS21	CO1	L2
	10.	Discuss various methods of collecting various data	2	SHORT	23AHS21	CO1	L2

	11.	Explain the precautions to be taken in using secondary data	2	SHORT	23AHS21	CO1	L2													
	12.	Explain the importance and significance of Sample.	2	SHORT	23AHS21	CO1	L2													
	13.	Calculate the arithmetic Mean of the following data. 40.50.55.78.58.60.	2	SHORT	23AHS21	CO1	L2													
	14.	Find the median from the given data. 57.58.61.42.55.90.	2	SHORT	23AHS21	CO1	L2													
	15.	Find the mode of the following data. 2,3,5,6,8,9,2,2,5.	2	SHORT	23AHS21	CO1	L2													
	10 Marks																			
	1.	Calculate the median from the following data	10	DESCRIPTIVE	23AHS21	CO1	L3													
		<table><tr><td>Marks</td><td>10-25</td><td>25-40</td><td>40-55</td><td>55-70</td><td>70-85</td><td>85-100</td></tr><tr><td>Frequency</td><td>6</td><td>20</td><td>44</td><td>26</td><td>3</td><td>1</td></tr></table>						Marks	10-25	25-40	40-55	55-70	70-85	85-100	Frequency	6	20	44	26	3
Marks		10-25						25-40	40-55	55-70	70-85	85-100								
Frequency	6	20	44	26	3	1														
	2.	Calculate the rank correlation coefficient for the following data	10	DESCRIPTIVE	23AHS21	CO1	L3													
	3.	Find the mode of the following distribution	10	DESCRIPTIVE	23AHS21															

		Class interval	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80				CO1	L3																						
		Frequency	5	8	7	12	28	20	10	10																											
4.	Obtain the rank correlation coefficient for the following data <table><tr><td>X</td><td>68</td><td>64</td><td>75</td><td>50</td><td>64</td><td>80</td><td>75</td><td>40</td><td>55</td><td>64</td></tr><tr><td>Y</td><td>62</td><td>58</td><td>68</td><td>45</td><td>81</td><td>60</td><td>68</td><td>48</td><td>50</td><td>70</td></tr></table>										X	68	64	75	50	64	80	75	40	55	64	Y	62	58	68	45	81	60	68	48	50	70	10	DESCRIPTIVE	23AHS21	CO1	L3
X	68	64	75	50	64	80	75	40	55	64																											
Y	62	58	68	45	81	60	68	48	50	70																											
5.	Calculate the variance and standard deviation of the following continuous frequency distribution <table><tr><td>Class interval</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td><td>70-80</td><td>80-90</td><td>90-100</td></tr><tr><td>Frequency</td><td>3</td><td>7</td><td>12</td><td>15</td><td>8</td><td>3</td><td>2</td></tr></table>										Class interval	30-40	40-50	50-60	60-70	70-80	80-90	90-100	Frequency	3	7	12	15	8	3	2	10	DESCRIPTIVE	23AHS21	CO1	L3						
Class interval	30-40	40-50	50-60	60-70	70-80	80-90	90-100																														
Frequency	3	7	12	15	8	3	2																														
6.	Calculate the regression equations of Y on X from the following data <table><tr><td>Price(Rs)</td><td>10</td><td>12</td><td>13</td><td>12</td><td>16</td><td>15</td></tr><tr><td>Amount Demanded</td><td>40</td><td>38</td><td>43</td><td>45</td><td>37</td><td>43</td></tr></table>										Price(Rs)	10	12	13	12	16	15	Amount Demanded	40	38	43	45	37	43	10	DESCRIPTIVE	23AHS21	CO1	L3								
Price(Rs)	10	12	13	12	16	15																															
Amount Demanded	40	38	43	45	37	43																															
7.	Calculate the coefficient of correlation for the following data <table><tr><td>X</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>Y</td><td>15</td><td>16</td><td>14</td><td>13</td><td>11</td><td>12</td><td>10</td><td>8</td><td>9</td></tr></table>										X	9	8	7	6	5	4	3	2	1	Y	15	16	14	13	11	12	10	8	9	10	DESCRIPTIVE	23AHS21	CO1	L3		
X	9	8	7	6	5	4	3	2	1																												
Y	15	16	14	13	11	12	10	8	9																												
8	Determine the two regression equations from the following data										10	DESCRIPTIVE																									

		<table><tr><td>X</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td><td>14</td></tr><tr><td>Y</td><td>4</td><td>2</td><td>5</td><td>10</td><td>4</td><td>11</td><td>12</td></tr></table>	X	2	4	6	8	10	12	14	Y	4	2	5	10	4	11	12			23AHS21	CO1	L3								
X	2	4	6	8	10	12	14																								
Y	4	2	5	10	4	11	12																								
9.	Determine the correlation coefficient between X and Y to the following data	<table><tr><td>X</td><td>5.5</td><td>3.6</td><td>2.6</td><td>3.4</td><td>3.1</td><td>2.7</td><td>3.0</td><td>2.0</td><td>2.3</td><td>3.1</td><td>3.2</td></tr><tr><td>Y</td><td>27</td><td>36</td><td>39</td><td>39</td><td>32</td><td>35</td><td>40</td><td>33</td><td>59</td><td>36</td><td>44</td></tr></table>	X	5.5	3.6	2.6	3.4	3.1	2.7	3.0	2.0	2.3	3.1	3.2	Y	27	36	39	39	32	35	40	33	59	36	44	10	DESCRIPTIVE	23AHS21	CO1	L3
X	5.5	3.6	2.6	3.4	3.1	2.7	3.0	2.0	2.3	3.1	3.2																				
Y	27	36	39	39	32	35	40	33	59	36	44																				
10.	From the following data, obtain two equations of regression lines (i) X on Y (ii) Y on X	<table><tr><td>X</td><td>6</td><td>2</td><td>10</td><td>4</td><td>8</td></tr><tr><td>Y</td><td>9</td><td>11</td><td>5</td><td>8</td><td>7</td></tr></table>	X	6	2	10	4	8	Y	9	11	5	8	7	10	DESCRIPTIVE	23AHS21	CO1	L3												
X	6	2	10	4	8																										
Y	9	11	5	8	7																										
11.	Obtain the two-regression line from the following data	<table><tr><td>X</td><td>68</td><td>64</td><td>75</td><td>50</td><td>64</td><td>80</td><td>75</td><td>40</td><td>55</td><td>64</td></tr><tr><td>Y</td><td>62</td><td>58</td><td>68</td><td>45</td><td>81</td><td>60</td><td>68</td><td>48</td><td>50</td><td>70</td></tr></table>	X	68	64	75	50	64	80	75	40	55	64	Y	62	58	68	45	81	60	68	48	50	70	10	DESCRIPTIVE	23AHS21	CO1	L3		
X	68	64	75	50	64	80	75	40	55	64																					
Y	62	58	68	45	81	60	68	48	50	70																					
12.	Find the variance and Standard deviation for the given Data. <table><tr><td>x</td><td>6</td><td>10</td><td>14</td><td>18</td><td>24</td><td>28</td><td>30</td></tr><tr><td>f</td><td>2</td><td>4</td><td>7</td><td>12</td><td>8</td><td>4</td><td>3</td></tr></table>	x	6	10	14	18	24	28	30	f	2	4	7	12	8	4	3	10	DESCRIPTIVE	23AHS21	CO1	L3									
x	6	10	14	18	24	28	30																								
f	2	4	7	12	8	4	3																								
13.	From the prices of shares X and Y given below for 10 days of trading which share is more stable? <table><tr><td>X</td><td>35</td><td>54</td><td>52</td><td>53</td><td>56</td><td>58</td><td>52</td><td>50</td><td>51</td><td>49</td></tr><tr><td>Y</td><td>108</td><td>107</td><td>105</td><td>105</td><td>106</td><td>107</td><td>104</td><td>103</td><td>104</td><td>101</td></tr></table>	X	35	54	52	53	56	58	52	50	51	49	Y	108	107	105	105	106	107	104	103	104	101	10	DESCRIPTIVE	23AHS21	CO1	L3			
X	35	54	52	53	56	58	52	50	51	49																					
Y	108	107	105	105	106	107	104	103	104	101																					
14.	From the following distribution calculate 1. First 4 moments about the mean. 2. Skewness based on moments. 3. Kurtosis <table><tr><td>Income(Rs)</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td></tr><tr><td>f</td><td>1</td><td>3</td><td>4</td><td>2</td></tr></table>	Income(Rs)	0-10	10-20	20-30	30-40	f	1	3	4	2	10	DESCRIPTIVE	23AHS21	CO1	L3															
Income(Rs)	0-10	10-20	20-30	30-40																											
f	1	3	4	2																											

UNIT-N0.2 Probability							
UNIT- N0.2	Q. No.	2 Marks	Marks	Question Type	Subject code	CO	BL
	1.	Define discrete and continuous random variables.	2	SHORT	23AHS21	CO2	L2
	2.	Find k if $f(x) = kx$, $0 < x < 1$ is a probability density function.	2	SHORT	23AHS21	CO2	L2
	3.	Find the maximum value of probability of an event.	2	SHORT	23AHS21	CO2	L2
	4.	Define discrete random variable with example.	2	SHORT	23AHS21	CO2	L2
	5.	Find k if $f(x) = kx$, $0 < x < 2$ is a probability density function.	2	SHORT	23AHS21	CO2	L2
	6.	State addition theorem on probability.	2	SHORT	23AHS21	CO2	L1
	7.	Define conditional probability.	2	SHORT	23AHS21	CO2	L2
	8.	Define continuous random variable with example.	2	SHORT	23AHS21	CO2	L2
	9.	Define Mathematical expectation.	2	SHORT	23AHS21	CO2	L2
	10.	Write the axioms of probability.	2	SHORT	23AHS21	CO2	L2
	11.	What is the probability that a card drawn at random from the pack of playing cards may be either a queen or a king.	2	SHORT	23AHS21	CO2	L2
	12.	A card is drawn from a well shuffled pack of cards. What is the probability that it is either a spade or an ace.	2	SHORT	23AHS21	CO2	L2
	13.	What is the probability of tossing two coins for head?	2	SHORT	23AHS21	CO2	L2
	14.	State Baye's theorem.	2	SHORT	23AHS21	CO2	
UNIT- N0.2	Q. No.	10 Marks	Mark s	Question Type	Subject code	CO	BL

	1.	A random variable X has the following probability function. Determine (i) k (ii) Mean (iii) Variance <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>$p(x)$</td><td>0</td><td>k</td><td>$2k$</td><td>$2k$</td><td>$3k$</td><td>k^2</td><td>$2k^2$</td><td>$7k^2 + k$</td></tr></table>	x	0	1	2	3	4	5	6	7	$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$	10	DESCRIPTIVE	23AHS21	CO2	L3
x	0	1	2	3	4	5	6	7																	
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$																	
	2.	The continuous probability function $f(x) = kx^2 e^{-x}$ when $x \geq 0$. Find (i) k (ii) Mean (iii) Variance.	10	DESCRIPTIVE	23AHS21	CO2	L3																		
	3.	A random variable X has the following probability distribution. Find the value of (i) k (ii) $P(X \leq 2)$ (iii) $P(2 \leq X \leq 5)$ <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>$p(x)$</td><td>0</td><td>$2k$</td><td>$3k$</td><td>$4k$</td><td>$5k$</td><td>$6k$</td><td>$7k$</td><td>$8k$</td></tr></table>	x	0	1	2	3	4	5	6	7	$p(x)$	0	$2k$	$3k$	$4k$	$5k$	$6k$	$7k$	$8k$	10	DESCRIPTIVE	23AHS21	CO2	L3
x	0	1	2	3	4	5	6	7																	
$p(x)$	0	$2k$	$3k$	$4k$	$5k$	$6k$	$7k$	$8k$																	
	4.	A random variable X has the following probability function. Determine (i) k (ii) Mean (iii) Variance <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>$p(x)$</td><td>0</td><td>k</td><td>$2k$</td><td>$2k$</td><td>$3k$</td><td>k^2</td><td>$2k^2$</td><td>$7k^2 + k$</td></tr></table>	x	0	1	2	3	4	5	6	7	$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$	10	DESCRIPTIVE	23AHS21	CO2	L3
x	0	1	2	3	4	5	6	7																	
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$																	
	5.	A bag contains 8 red and 6 blue balls. Two drawing of each 2 balls are made. Find the probability that the first drawing gives two red balls and second	10	DESCRIPTIVE	23AHS21																				

		drawing gives 2 blue balls,if the balls drawn first are replaced before the second draw.				CO2	L3
6.		Three machines I,II,III produce 40%,30%,30% of the total number of items of factory.The percentages of defective items of these machines are 4%,2%,3%.If an item is selected at random,find the probability that the item is defective.	10	DESCRIPTIVE	23AHS21	CO2	L3
7.		The probabilities that students A,B,C,D solve a problem are $\frac{1}{3}, \frac{2}{5}, \frac{1}{5}$ and $\frac{1}{4}$ respectively.If all of them try to solve the problem,what is the probability that the problem is solved.	10	DESCRIPTIVE	23AHS21	CO2	L3
8.		In a bolt factory machines A,B,C manufacture 20%,30% and 50% of the total of their output and 6%,3% and 2% are defective.A bolt is drawn at random and found to be defective.Find the probability that it is manufactured from (i)Machine A.(ii)Machine C.	10	DESCRIPTIVE	23AHS21	CO2	L3
9.		First box contains 2 black,3 red,1 white balls;second box contains 1 black,1 red,2 white balls and third box contains 5 black,3 red,4 white balls.Of these a box is selected at random.From it a red ball is randomly drawn.If the ball is red,find the probability that it is from second box.	10	DESCRIPTIVE	23AHS21	CO2	L3
10.		A businessman goes to hotels X,Y,Z,20%,50%,30% of the time respectively.It is known that 5%,4%,8% of the rooms in X,Y,Z hotels have faulty plumbings.What is the probability that business man's room having faulty piumbing is assigned to hotel Z.	10	DESCRIPTIVE	23AHS21	CO2	L3
11.		For the continuous Random Variable X whose probability density function is given by $f(x) = cx(2 - x)$, if $0 \leq x \leq 2$ where c is a constant 1. Find C 2. Mean and Variance.	10	DESCRIPTIVE	23AHS21	CO2	L3
12.		Suppose a continuous Random Variable X has the probability density $f(x) = k(1 - x^2)$, $0 < x < 1$ and $f(x) = 0$, Otherwise Find 1. Mean 2. Variance	10	DESCRIPTIVE	23AHS21	CO2	L3
13.		Find the constant k such that $f(x) = kx^2$, $0 < x < 3$ is a probability function. Find the distribution function F(x), $P(1 < X \leq 2)$.	10	DESCRIPTIVE	23AHS21	CO2	L3

	14.	A continuous Random variable has the probability density function $f(x) = kxe^{-\lambda x}, x \geq 0, \lambda \geq 0$ Determine k, mean, variance	10	DESCRIPTIVE	23AHS21	CO2	L3
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